

power. Note that in the given problem, the first series starts with  $x^0$ , whereas the second series starts with  $x^1$ . By writing the first term of the first series outside of the summation notation,

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1} = 2 \cdot 1 c_2 x^0 + \sum_{n=3}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1},$$

series starts with  $x$  for  $n = 3$ 
series starts with  $x$  for  $n = 0$

↓
↓

we see that both series on the right side start with the same power of  $x$ ; namely,  $x^1$ . Now to get the same summation index we are inspired by the exponents of  $x$ ; we let  $k = n - 2$  in the first series and at the same time let  $k = n + 1$  in the second series. The right side becomes

$$2c_2 + \sum_{k=1}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} c_{k-1} x^k. \quad (3)$$

↑
↑

same
same

Remember, the summation index is a “dummy” variable; the fact that  $k = n - 1$  in one case and  $k = n + 1$  in the other should cause no confusion if you keep in mind that it is the *value* of the summation index that is important. In both cases  $k$  takes on the same successive values  $k = 1, 2, 3, \dots$  when  $n$  takes on the values  $n = 2, 3, 4, \dots$  for  $k = n - 1$  and  $n = 0, 1, 2, \dots$  for  $k = n + 1$ . We are now in a position to add the series in (3) term by term:

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1} = 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + c_{k-1}] x^k. \quad (4) \equiv$$

If you are not convinced of the result in (4), then write out a few terms on both sides of the equality.

## 5.1.2 Power Series Solutions

**A Definition** Suppose the linear second-order differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \quad (5)$$

is put into standard form

$$y'' + P(x)y' + Q(x)y = 0 \quad (6)$$

by dividing by the leading coefficient  $a_2(x)$ . We make the following definition.

### Definition 5.1.1 Ordinary and Singular Points

A point  $x_0$  is said to be an **ordinary point** of the differential equation (5) if both  $P(x)$  and  $Q(x)$  in the standard form (6) are analytic at  $x_0$ . A point that is not an ordinary point is said to be a **singular point** of the equation.

Every finite value of  $x$  is an ordinary point of  $y'' + (e^x)y' + (\sin x)y = 0$ . In particular,  $x = 0$  is an ordinary point, since, as we have already seen in (2), both  $e^x$  and  $\sin x$  are analytic at this point. The negation in the second sentence of Definition 5.1.1 stipulates that if at least one of the functions  $P(x)$  and  $Q(x)$  in (6) fails to be analytic at  $x_0$ , then  $x_0$  is a singular point. Note that  $x = 0$  is a singular point of the differential equation  $y'' + (e^x)y' + (\ln x)y = 0$ , since  $Q(x) = \ln x$  is discontinuous at  $x = 0$  and so cannot be represented by a power series in  $x$ .

**Polynomial Coefficients** We shall be interested primarily in the case in which (5) has polynomial coefficients. A polynomial is analytic at any value  $x$ , and a rational function is analytic *except* at points where its denominator is zero. Thus, if  $a_2(x)$ ,  $a_1(x)$ , and  $a_0(x)$  are polynomials with no common factors, then both rational functions  $P(x) = a_1(x)/a_2(x)$  and  $Q(x) = a_0(x)/a_2(x)$  are analytic except where  $a_2(x) = 0$ .

*It follows, then, that  $x = x_0$  is an ordinary point of (5) if  $a_2(x_0) \neq 0$ , whereas  $x = x_0$  is a singular point of (5) if  $a_2(x_0) = 0$ .*

For example, the only singular points of the equation  $(x^2 - 1)y'' + 2xy' + 6y = 0$  are solutions of  $x^2 - 1 = 0$  or  $x = \pm 1$ . All other finite values\* of  $x$  are ordinary points. Inspection of the Cauchy–Euler equation  $ax^2y'' + bxy' + cy = 0$  shows that it has a singular point at  $x = 0$ . Singular points need not be real numbers. The equation  $(x^2 + 1)y'' + xy' - y = 0$  has singular points at the solutions of  $x^2 + 1 = 0$ ; namely,  $x = \pm i$ . All other values of  $x$ , real or complex, are ordinary points.

We state the following theorem about the existence of power series solutions without proof.

### Theorem 5.1.1 Existence of Power Series Solutions

If  $x = x_0$  is an ordinary point of the differential equation (5), we can always find two linearly independent solutions in the form of a power series centered at  $x_0$ ; that is,  $y = \sum_{n=0}^{\infty} c_n(x - x_0)^n$ . A series solution converges at least on some interval defined by  $|x - x_0| < R$ , where  $R$  is the distance from  $x_0$  to the closest singular point.

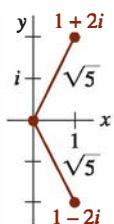


FIGURE 5.1.2 Distance from ordinary point 0 to singular points

All the power series solutions will be centered at 0.

A solution of the form  $y = \sum_{n=0}^{\infty} c_n(x - x_0)^n$  is said to be a **solution about the ordinary point  $x_0$** . The distance  $R$  in Theorem 5.1.1 is the *minimum* value or **lower bound** for the radius of convergence. For example, the complex numbers  $1 \pm 2i$  are singular points of  $(x^2 - 2x + 5)y'' + xy' - y = 0$ , but since  $x = 0$  is an ordinary point of the equation, Theorem 5.1.1 guarantees that we can find two power series solutions centered at 0. That is, the solutions look like  $y = \sum_{n=0}^{\infty} c_n x^n$ , and, moreover, we know without actually finding these solutions that each series must converge *at least* for  $|x| < \sqrt{5}$ , where  $R = \sqrt{5}$  is the distance in the complex plane from 0 to either of the numbers  $1 + 2i$  or  $1 - 2i$ . See FIGURE 5.1.2. However, the differential equation has a solution that is valid for much larger values of  $x$ ; indeed, this solution is valid on the interval  $(-\infty, \infty)$  because it can be shown that one of the two solutions is a polynomial.

In the examples that follow, as well as in Exercises 5.1, we shall, for the sake of simplicity, only find power series solutions about the ordinary point  $x = 0$ . If it is necessary to find a power series solution of an ODE about an ordinary point  $x_0 \neq 0$ , we can simply make the change of variable  $t = x - x_0$  in the equation (this translates  $x = x_0$  to  $t = 0$ ), find solutions of the new equation of the form  $y = \sum_{n=0}^{\infty} c_n t^n$ , and then resubstitute  $t = x - x_0$ .

Finding a power series solution of a homogeneous linear second-order ODE has been accurately described as “the method of undetermined *series* coefficients,” since the procedure is quite analogous to what we did in Section 3.4. In brief, here is the idea: We substitute  $y = \sum_{n=0}^{\infty} c_n x^n$  into the differential equation, combine series as we did in Example 1, and then equate all coefficients to the right side of the equation to determine the coefficients  $c_n$ . But since the right side is zero, the last step requires, by the identity property in the preceding bulleted list, that all coefficients of  $x$  must be equated to zero. No, this does *not* mean that all coefficients *are* zero; this would not make sense, because after all, Theorem 5.1.1 guarantees that we can find two linearly independent solutions. Example 2 illustrates how the single assumption that  $y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \dots$  leads to two sets of coefficients so that we have two distinct power series  $y_1(x)$  and  $y_2(x)$ , both expanded about the ordinary point  $x = 0$ . The general solution of the differential equation is  $y = C_1 y_1(x) + C_2 y_2(x)$ ; indeed, if  $y_1(0) = 1$ ,  $y_1'(0) = 0$ , and  $y_2(0) = 0$ ,  $y_2'(0) = 1$ , then  $C_1 = c_0$  and  $C_2 = c_1$ .

### EXAMPLE 2 Power Series Solutions

Solve  $y'' + xy = 0$ .

**SOLUTION** Since there are no finite singular points, Theorem 5.1.1 guarantees two power series solutions, centered at 0, convergent for  $|x| < \infty$ . Substituting  $y = \sum_{n=0}^{\infty} c_n x^n$  and the second derivative  $y'' = \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}$  (see (1)) into the differential equation gives

$$y'' + xy = \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + x \sum_{n=0}^{\infty} c_n x^n = \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1}. \quad (7)$$

\*For our purposes, ordinary points and singular points will always be finite points. It is possible for an ODE to have, say, a singular point at infinity.

Now we have already added the last two series on the right side of the equality in (7) by shifting the summation index in Example 1. From the result given in (4),

$$y'' + xy = 2c_2 + \sum_{k=1}^{\infty} [(k+1)(k+2)c_{k+2} + c_{k-1}]x^k = 0. \quad (8)$$

At this point we invoke the identity property. Since (8) is identically zero, it is necessary that the coefficient of each power of  $x$  be set equal to zero; that is,  $2c_2 = 0$  (it is the coefficient of  $x^0$ ), and

$$(k+1)(k+2)c_{k+2} + c_{k-1} = 0, \quad k = 1, 2, 3, \dots \quad (9)$$

Now  $2c_2 = 0$  obviously dictates that  $c_2 = 0$ . But the expression in (9), called a **recurrence relation**, determines the  $c_k$  in such a manner that we can choose a certain subset of the set of coefficients to be *nonzero*. Since  $(k+1)(k+2) \neq 0$  for all values of  $k$ , we can solve (9) for  $c_{k+2}$  in terms of  $c_{k-1}$ :

$$c_{k+2} = -\frac{c_{k-1}}{(k+1)(k+2)}, \quad k = 1, 2, 3, \dots \quad (10)$$

◀ This formula is called a two-term recurrence relation.

This relation generates consecutive coefficients of the assumed solution one at a time as we let  $k$  take on the successive integers indicated in (10):

$$\begin{aligned} k=1, \quad c_3 &= -\frac{c_0}{2 \cdot 3} \\ k=2, \quad c_4 &= -\frac{c_1}{3 \cdot 4} \\ k=3, \quad c_5 &= -\frac{c_2}{4 \cdot 5} = 0 \quad \leftarrow c_2 \text{ is zero} \\ k=4, \quad c_6 &= -\frac{c_3}{5 \cdot 6} = \frac{1}{2 \cdot 3 \cdot 5 \cdot 6} c_0 \\ k=5, \quad c_7 &= -\frac{c_4}{6 \cdot 7} = \frac{1}{3 \cdot 4 \cdot 6 \cdot 7} c_1 \\ k=6, \quad c_8 &= -\frac{c_5}{7 \cdot 8} = 0 \quad \leftarrow c_5 \text{ is zero} \\ k=7, \quad c_9 &= -\frac{c_6}{8 \cdot 9} = -\frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9} c_0 \\ k=8, \quad c_{10} &= -\frac{c_7}{9 \cdot 10} = -\frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10} c_1 \\ k=9, \quad c_{11} &= -\frac{c_8}{10 \cdot 11} = 0 \quad \leftarrow c_8 \text{ is zero} \end{aligned}$$

and so on. Now substituting the coefficients just obtained into the original assumption

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + c_{11}x^{11} + \dots$$

we get

$$\begin{aligned} y &= c_0 + c_1x + 0 - \frac{c_0}{2 \cdot 3}x^3 - \frac{c_1}{3 \cdot 4}x^4 + 0 + \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6}x^6 \\ &\quad + \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7}x^7 + 0 - \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9}x^9 - \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10}x^{10} + 0 + \dots \end{aligned}$$

After grouping the terms containing  $c_0$  and the terms containing  $c_1$ , we obtain  $y = c_0y_1(x) + c_1y_2(x)$ , where

$$y_1(x) = 1 - \frac{1}{2 \cdot 3} x^3 + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6} x^6 - \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9} x^9 + \cdots = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{2 \cdot 3 \cdots (3k-1)(3k)} x^{3k}$$

$$y_2(x) = x - \frac{1}{3 \cdot 4} x^4 + \frac{1}{3 \cdot 4 \cdot 6 \cdot 7} x^7 - \frac{1}{3 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 10} x^{10} + \cdots = x + \sum_{k=1}^{\infty} \frac{(-1)^k}{3 \cdot 4 \cdots (3k)(3k+1)} x^{3k+1}.$$

Since the recursive use of (10) leaves  $c_0$  and  $c_1$  completely undetermined, they can be chosen arbitrarily. As mentioned prior to this example, the linear combination  $y = c_0 y_1(x) + c_1 y_2(x)$  actually represents the general solution of the differential equation. Although we know from Theorem 5.1.1 that each series solution converges for  $|x| < \infty$ , this fact can also be verified by the ratio test.  $\equiv$

The differential equation in Example 2 is called **Airy's equation** and is named after the English astronomer and mathematician **Sir George Biddell Airy** (1801–1892). Airy's equation is encountered in the study of diffraction of light, diffraction of radio waves around the surface of the Earth, aerodynamics, and the deflection of a uniform thin vertical column that bends under its own weight. Other common forms of Airy's equation are  $y'' - xy = 0$  and  $y'' + \alpha^2 xy = 0$ . See Problem 44 in Exercises 5.3 for an application of the last equation. It should also be apparent by making a sign change in (10) of Example 2 that the general solution of  $y'' - xy = 0$  is  $y = c_0 y_1(x) + c_1 y_2(x)$ , where the series solutions are in this case

$$y_1(x) = 1 + \frac{1}{2 \cdot 3} x^3 + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6} x^6 + \frac{1}{2 \cdot 3 \cdot 5 \cdot 6 \cdot 8 \cdot 9} x^9 + \cdots = 1 + \sum_{k=1}^{\infty} \frac{1}{2 \cdot 3 \cdots (3k-1)(3k)} x^{3k}$$

$$y_2(x) = x + \frac{1}{3 \cdot 4} x^4 + \frac{1}{3 \cdot 4 \cdot 6 \cdot 7} x^7 + \frac{1}{2 \cdot 3 \cdot 6 \cdot 7 \cdot 9 \cdot 10} x^{10} + \cdots = x + \sum_{k=1}^{\infty} \frac{1}{3 \cdot 4 \cdots (3k)(3k+1)} x^{3k+1}.$$

### EXAMPLE 3 Power Series Solution

Solve  $(x^2 + 1)y'' + xy' - y = 0$ .

**SOLUTION** As we have already seen on page 258, the given differential equation has singular points at  $x = \pm i$ , and so a power series solution centered at 0 will converge at least for  $|x| < 1$ , where 1 is the distance in the complex plane from 0 to either  $i$  or  $-i$ . The assumption  $y = \sum_{n=0}^{\infty} c_n x^n$  and its first two derivatives (see (1)) lead to

$$\begin{aligned} (x^2 + 1) \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x \sum_{n=1}^{\infty} n c_n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n \\ &= \sum_{n=2}^{\infty} n(n-1)c_n x^n + \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} n c_n x^n - \sum_{n=0}^{\infty} c_n x^n \\ &= 2c_2 x^0 - c_0 x^0 + 6c_3 x + c_1 x - c_1 x + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} \\ &\quad + \underbrace{\sum_{n=4}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=2}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=2}^{\infty} c_n x^n}_{k=n} \\ &= 2c_2 - c_0 + 6c_3 x + \sum_{k=2}^{\infty} [k(k-1)c_k + (k+2)(k+1)c_{k+2} + kc_k - c_k] x^k \\ &= 2c_2 - c_0 + 6c_3 x + \sum_{k=2}^{\infty} [(k+1)(k-1)c_k + (k+2)(k+1)c_{k+2}] x^k = 0. \end{aligned}$$

From this last identity we conclude that  $2c_2 - c_0 = 0$ ,  $6c_3 = 0$ , and

$$(k+1)(k-1)c_k + (k+2)(k+1)c_{k+2} = 0.$$

Thus, 
$$c_2 = \frac{1}{2}c_0, \quad c_3 = 0,$$

$$c_{k+2} = \frac{1-k}{k+2}c_k, \quad k = 2, 3, 4, \dots$$

Substituting  $k = 2, 3, 4, \dots$  into the last formula gives

$$c_4 = -\frac{1}{4}c_2 = -\frac{1}{2 \cdot 4}c_0 = -\frac{1}{2^2 2!}c_0$$

$$c_5 = -\frac{2}{5}c_3 = 0 \quad \leftarrow c_3 \text{ is zero}$$

$$c_6 = -\frac{3}{6}c_4 = \frac{3}{2 \cdot 4 \cdot 6}c_0 = \frac{1 \cdot 3}{2^3 3!}c_0$$

$$c_7 = -\frac{4}{7}c_5 = 0 \quad \leftarrow c_5 \text{ is zero}$$

$$c_8 = -\frac{5}{8}c_6 = -\frac{3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 8}c_0 = -\frac{1 \cdot 3 \cdot 5}{2^4 4!}c_0$$

$$c_9 = -\frac{6}{9}c_7 = 0 \quad \leftarrow c_7 \text{ is zero}$$

$$c_{10} = -\frac{7}{10}c_8 = \frac{3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10}c_0 = -\frac{1 \cdot 3 \cdot 5 \cdot 7}{2^5 5!}c_0$$

and so on. Therefore,

$$\begin{aligned} y &= c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + c_9x^9 + c_{10}x^{10} + \dots \\ &= c_0 \left[ 1 + \frac{1}{2}x^2 - \frac{1}{2^2 2!}x^4 + \frac{1 \cdot 3}{2^3 3!}x^6 - \frac{1 \cdot 3 \cdot 5}{2^4 4!}x^8 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{2^5 5!}x^{10} - \dots \right] + c_1x \\ &= c_0 y_1(x) + c_1 y_2(x). \end{aligned}$$

The solutions are the polynomial  $y_2(x) = x$  and the power series

$$y_1(x) = 1 + \frac{1}{2}x^2 + \sum_{n=2}^{\infty} (-1)^{n-1} \frac{1 \cdot 3 \cdot 5 \cdots (2n-3)}{2^n n!} x^{2n}, \quad |x| < 1. \quad \equiv$$

#### EXAMPLE 4 Three-Term Recurrence Relation

If we seek a power series solution  $y = \sum_{n=0}^{\infty} c_n x^n$  for the differential equation

$$y'' - (1+x)y = 0,$$

we obtain  $c_2 = c_0/2$  and the recurrence relation

$$c_{k+2} = \frac{c_k + c_{k-1}}{(k+1)(k+2)}, \quad k = 1, 2, 3, \dots$$

◀ This formula is called a three-term recurrence relation.

Examination of the formula shows that the coefficients  $c_3, c_4, c_5, \dots$  are expressed in terms of *both*  $c_1$  and  $c_0$ , and moreover, the algebra required to do this becomes a little messy. To simplify life we can first choose  $c_0 \neq 0, c_1 = 0$ ; this yields consecutive coefficients for one solution that are expressed entirely in terms of  $c_0$ . If we next choose  $c_0 = 0, c_1 \neq 0$ , then the

coefficients for the other solution are expressed in terms of  $c_1$ . Using  $c_2 = \frac{1}{2}c_0$  in both cases, the recurrence relation for  $k = 1, 2, 3, \dots$  gives

$$c_0 \neq 0, c_1 = 0$$

$$c_2 = \frac{1}{2}c_0$$

$$c_3 = \frac{c_1 + c_0}{2 \cdot 3} = \frac{c_0}{2 \cdot 3} = \frac{1}{6}c_0$$

$$c_4 = \frac{c_2 + c_1}{3 \cdot 4} = \frac{c_0}{2 \cdot 3 \cdot 4} = \frac{1}{24}c_0$$

$$c_5 = \frac{c_3 + c_2}{4 \cdot 5} = \frac{c_0}{4 \cdot 5} \left[ \frac{1}{6} + \frac{1}{2} \right] = \frac{1}{30}c_0$$

$$c_0 = 0, c_1 \neq 0$$

$$c_2 = \frac{1}{2}c_0 = 0$$

$$c_3 = \frac{c_1 + c_0}{2 \cdot 3} = \frac{c_1}{2 \cdot 3} = \frac{1}{6}c_1$$

$$c_4 = \frac{c_2 + c_1}{3 \cdot 4} = \frac{c_1}{3 \cdot 4} = \frac{1}{12}c_1$$

$$c_5 = \frac{c_3 + c_2}{4 \cdot 5} = \frac{c_1}{4 \cdot 5 \cdot 6} = \frac{1}{120}c_1$$

and so on. Finally, we see that the general solution of the equation is  $y = c_0y_1(x) + c_1y_2(x)$ , where

$$y_1(x) = 1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{30}x^5 + \dots$$

$$\text{and } y_2(x) = x + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{120}x^5 + \dots$$

Each series converges for all finite values of  $x$ . ≡

□ **Nonpolynomial Coefficients** The next example illustrates how to find a power series solution about the ordinary point  $x_0 = 0$  of a differential equation when its coefficients are not polynomials. In this example we see an application of multiplication of two power series.

#### EXAMPLE 5 ODE with Nonpolynomial Coefficients

Solve  $y'' + (\cos x)y = 0$ .

**SOLUTION** We see  $x = 0$  is an ordinary point of the equation because, as we have already seen,  $\cos x$  is analytic at that point. Using the Maclaurin series for  $\cos x$  given in (2), along with the usual assumption  $y = \sum_{n=0}^{\infty} c_n x^n$  and the results in (1), we find

$$\begin{aligned} y'' + (\cos x)y &= \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right) \sum_{n=0}^{\infty} c_n x^n \\ &= 2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + \dots + \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots\right)(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots) \\ &= 2c_2 + c_0 + (6c_3 + c_1)x + \left(12c_4 + c_2 - \frac{1}{2}c_0\right)x^2 + \left(20c_5 + c_3 - \frac{1}{2}c_1\right)x^3 + \dots = 0. \end{aligned}$$

It follows that

$$2c_2 + c_0 = 0, \quad 6c_3 + c_1 = 0, \quad 12c_4 + c_2 - \frac{1}{2}c_0 = 0, \quad 20c_5 + c_3 - \frac{1}{2}c_1 = 0,$$

and so on. This gives  $c_2 = -\frac{1}{2}c_0$ ,  $c_3 = -\frac{1}{6}c_1$ ,  $c_4 = \frac{1}{12}c_0$ ,  $c_5 = \frac{1}{30}c_1, \dots$ . By grouping terms we arrive at the general solution  $y = c_0y_1(x) + c_1y_2(x)$ , where

$$y_1(x) = 1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \dots \quad \text{and} \quad y_2(x) = x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \dots$$

Since the differential equation has no finite singular points, both power series converge for  $|x| < \infty$ . ≡

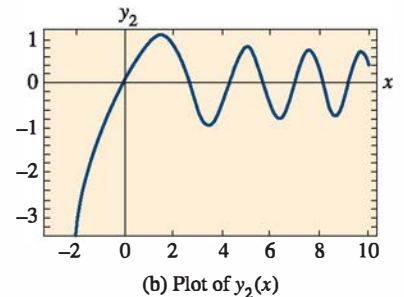
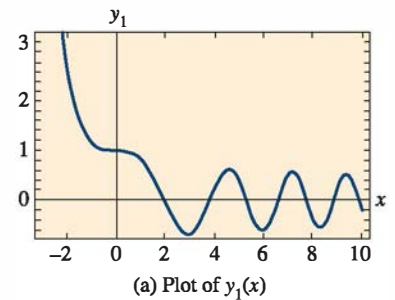
**□ Solution Curves** The approximate graph of a power series solution  $y(x) = \sum_{n=0}^{\infty} c_n x^n$  can be obtained in several ways. We can always resort to graphing the terms in the sequence of partial sums of the series; in other words, the graphs of the polynomials  $S_N(x) = \sum_{n=0}^N c_n x^n$ . For large values of  $N$ ,  $S_N(x)$  should give us an indication of the behavior of  $y(x)$  near the ordinary point  $x = 0$ . We can also obtain an approximate solution curve by using a numerical solver as we did in Section 3.11. For example, if you carefully scrutinize the series solutions of Airy's equation in Example 2, you should see that  $y_1(x)$  and  $y_2(x)$  are, in turn, the solutions of the initial-value problems

$$\begin{aligned} y'' + xy &= 0, & y(0) &= 1, & y'(0) &= 0, \\ y'' + xy &= 0, & y(0) &= 0, & y'(0) &= 1. \end{aligned} \quad (11)$$

The specified initial conditions “pick out” the solutions  $y_1(x)$  and  $y_2(x)$  from  $y = c_0 y_1(x) + c_1 y_2(x)$ , since it should be apparent from our basic series assumption  $y = \sum_{n=0}^{\infty} c_n x^n$  that  $y(0) = c_0$  and  $y'(0) = c_1$ . Now if your numerical solver requires a system of equations, the substitution  $y' = u$  in  $y'' + xy = 0$  gives  $y'' = u' = -xy$ , and so a system of two first-order equations equivalent to Airy's equation is

$$\begin{aligned} y' &= u \\ u' &= -xy. \end{aligned} \quad (12)$$

Initial conditions for the system in (12) are the two sets of initial conditions in (11) but rewritten as  $y(0) = 1, u(0) = 0$ , and  $y(0) = 0, u(0) = 1$ . The graphs of  $y_1(x)$  and  $y_2(x)$  shown in **FIGURE 5.1.3** were obtained with the aid of a numerical solver using the fourth-order Runge–Kutta method with a step size of  $h = 0.1$ .



**FIGURE 5.1.3** Solutions of Airy's equation

## Remarks

- (i) In the problems that follow, do not expect to be able to write a solution in terms of summation notation in each case. Even though we can generate as many terms as desired in a series solution  $y = \sum_{n=0}^{\infty} c_n x^n$  either through the use of a recurrence relation or, as in Example 5, by multiplication, it may not be possible to deduce any general term for the coefficients  $c_n$ . We may have to settle, as we did in Examples 4 and 5, for just writing out the first few terms of the series.
- (ii) A point  $x_0$  is an ordinary point of a *nonhomogeneous* linear second-order DE  $y'' + P(x)y' + Q(x)y = f(x)$  if  $P(x)$ ,  $Q(x)$ , and  $f(x)$  are analytic at  $x_0$ . Moreover, Theorem 5.1.1 extends to such DEs—in other words, we can find power series solutions  $y = \sum_{n=0}^{\infty} c_n (x - x_0)^n$  of nonhomogeneous linear DEs in the same manner as in Examples 2–5. See Problem 36 in Exercises 5.1.

## 5.1 Exercises

Answers to selected odd-numbered problems begin on page ANS-11.

### 5.1.1 Review of Power Series

In Problems 1–4, find the radius of convergence and interval of convergence for the given power series.

- $\sum_{n=1}^{\infty} \frac{2^n}{n} x^n$
- $\sum_{n=0}^{\infty} \frac{(100)^n}{n!} (x + 7)^n$
- $\sum_{k=1}^{\infty} \frac{(-1)^k}{10^k} (x - 5)^k$
- $\sum_{k=0}^{\infty} k! (x - 1)^k$

In Problems 5 and 6, the given function is analytic at  $x = 0$ . Find the first four terms of a power series in  $x$ . Perform the multiplication by hand or use a CAS, as instructed.

- $\sin x \cos x$
- $e^{-x} \cos x$

In Problems 7 and 8, the given function is analytic at  $x = 0$ . Find the first four terms of a power series in  $x$ . Perform the long division by hand or use a CAS, as instructed. Give the open interval of convergence.

- $\frac{1}{\cos x}$
- $\frac{1 - x}{2 + x}$

In Problems 9 and 10, rewrite the given power series so that its general term involves  $x^k$ .

- $\sum_{n=1}^{\infty} n c_n x^{n+2}$
- $\sum_{n=3}^{\infty} (2n - 1) c_n x^{n-3}$